

WRO 2026 Robot Parts Cost and Selection Rationale

Official parts summary with selected quantities, prices, totals, and clear selection reasons.

1. Required Parts Cost Summary

Part Name	Qty	Unit Price	Total
4-Wheel RC Car Chassis Kit with Wheels and Battery Mount	1	350 NIS	350 NIS
ESP32 DevKit V1 Development Board	1	30 NIS	30 NIS
L298N Dual H-Bridge DC Motor Driver Module	1	10 NIS	10 NIS
TCRT5000 IR Reflective Sensor Module	1	5 NIS	5 NIS
VL53L0X Time-of-Flight Distance Sensor Module	2	15 NIS	30 NIS
TCS34725 RGB Color Sensor Module	2	40 NIS	80 NIS
9V Battery Pack / Robot Power Battery	1	30 NIS	30 NIS
ESP32 GPIO Expansion Shield	1	20 NIS	20 NIS
SPST ON/OFF Rocker Switch	1	10 NIS	10 NIS
MPU-6050 6-Axis Gyroscope and Accelerometer IMU Module	1	Unknown	-
Dupont Jumper Wire Kit (Male-Male, Male-Female, Female-Female)	As needed	Unknown	-

Known Required Total: 565 NIS

2. Optional Parts

Optional Part Name	Qty	Unit Price	Total
Raspberry Pi 4 Model B	1	800 NIS	800 NIS
Raspberry Pi 5	1	1000 NIS	1000 NIS
Raspberry Pi Camera Module / USB Camera	1	Unknown	-

Total with Raspberry Pi 4: 1,365 NIS + unknown camera, gyro, and wires costs

Total with Raspberry Pi 5: 1,565 NIS + unknown camera, gyro, and wires costs

3. Why Each Part Was Selected

Part Name	Purpose	Selection Reason
4-Wheel RC Car Chassis Kit with Wheels and Battery Mount	Ready base for the robot	It already includes wheels, motor mounting area, and battery/power space, so the team can focus on sensors, wiring, and control.
ESP32 DevKit V1 Development Board	Main controller for sensors and motors	It is low-cost, compact, and has enough GPIO pins for IR, color sensors, ToF sensors, gyro, and motor driver control.
L298N Dual H-Bridge DC Motor Driver Module	Controls DC motor direction and speed	It allows forward/reverse movement and PWM speed control for the chassis motors.
TCRT5000 IR Reflective Sensor Module	Line/edge reference sensing	It can detect contrast on the field surface and support simple line or boundary checking.
VL53L0X Time-of-Flight Distance Sensor Module	Short-range distance measurement	Two sensors help detect nearby obstacles or side distances more accurately than one sensor.
TCS34725 RGB Color Sensor Module	Color detection	Two color sensors give better left/right color reading and help detect field color markers more reliably.
9V Battery Pack / Robot Power Battery	Robot power source	It provides portable power for the robot during testing and movement.
ESP32 GPIO Expansion Shield	Cleaner wiring	It organizes VCC, GND, and signal pins, reducing wiring mistakes and making sensor connections easier.
SPST ON/OFF Rocker Switch	Safe power control	It lets the team turn the robot on and off quickly without disconnecting wires.
MPU-6050 6-Axis Gyroscope and Accelerometer IMU Module	Orientation and turning accuracy	It measures rotation and tilt, helping the robot drive straighter and make more accurate turns.
Dupont Jumper Wire Kit	Electrical connections	Needed to connect the ESP32, sensors, motor driver, shield, switch, and power parts.
Raspberry Pi 4 Model B / Raspberry Pi 5	Optional vision processor	Used only if the team decides to run camera-based lane or object detection; otherwise ESP32 can handle the basic sensor system.
Raspberry Pi Camera Module / USB Camera	Optional camera input	Needed only with Raspberry Pi if the robot uses computer vision.